

Water Consumption and the Unaccounted Cost of AI Cooling

Executive summary

This report finds that the water question for data centres in the UK, Ireland and the EU is not best understood as a single Europe-wide average. It is a local infrastructure and catchment question. The strongest evidence does **not** support the claim that all data centres are inherently major water users. In England, a 2025 techUK and Environment Agency-backed survey of 73 facilities found that 51% used waterless cooling systems, 64% used less than 10,000 m³ a year, and 89% either measured water use or used systems that do not require water for cooling. At the same time, the Environment Agency's 2025 national water-resources appendix found that data-centre water consumption in England was **underestimated** at about **1,125 megalitres a year** in 2024–25, that **83%** of observed consumption sat in the already water-scarce **Water Resources South East** region, and that summer peaks track temperature closely. In other words: the national picture can look small while the **regional and seasonal** picture looks materially important. ¹

The clearest site-level disclosures show very wide variation by design, climate, cooling system and disclosure quality. Meta's Clonee campus in Ireland withdrew **928 ML** in 2021, **659 ML** in 2023 and **571 ML** in 2024. Google's Dublin data centre, by contrast, was reported by An Fóram Uisce using Google's own environmental reporting as withdrawing only **0.6 million gallons** in 2023, about **2.7 million litres** for the year, because it had moved to an air-cooling configuration. In the Nordics, Meta reported **29 ML** at Luleå in 2024 and **292 ML** at Odense in 2024, while Google's Hamina site uses seawater cooling and off-site heat recovery rather than freshwater-based cooling. These figures do **not** prove that one operator is "good" and another "bad"; they show that **cooling architecture** and disclosure scope matter more than broad slogans about the cloud. ²

AI changes the water question, but not in a simple one-directional way. Higher rack densities, GPU clusters and AI training loads can push up heat rejection and therefore cooling intensity. Google explicitly said in its 2024 environmental reporting that growth in AI products and services increased data-centre workloads and the associated water footprint required to cool them efficiently. Yet AI-era cooling can also **reduce direct on-site water** in some designs: Microsoft says its new data centres designed for AI workloads are optimised to consume **zero water for cooling**, and that direct-to-chip cooling can save **more than 125 million litres per facility each year**. Google argues that water cooling can reduce energy use and related carbon emissions relative to many air-cooled designs, illustrating a real trade-off: lower **direct** water use may come with higher electricity demand and therefore higher **indirect** water use in the power system. Secondary analyses drawing on Lawrence Berkeley National Laboratory work suggest that indirect water from electricity can exceed direct cooling water by a large multiple, but site-specific European evidence remains sparse. ³

On regulation, the evidence is stronger on **transparency reform** than on **water allocation reform**. The EU's recast Energy Efficiency Directive and the 2024 delegated regulation create a public sustainability reporting scheme for data centres that includes **total water input** and **potable water input**. ESRs E3 under the CSRD framework requires undertakings to disclose material impacts, risks and opportunities relating to water and marine resources, including water consumption where material. Those are meaningful advances. But they are still mostly **reporting frameworks**, not hard-edged rules for

cumulative local water allocation under drought stress. The Environment Agency has already warned that future non-household demand forecasts are unlikely to capture future data-centre needs because of the speed of growth and lack of information on where and when sites will be built and which cooling systems they will use. That is the core policy gap: **permitting and planning still tend to see one site at a time, while water systems experience cumulative pressure at catchment scale and at peak heat moments.** ⁴

The main conclusion is therefore balanced but firm. Data centres are not automatically a water crisis. Some are genuinely low-water or near-zero-water in direct operation. Some use non-potable sources, seawater, rainwater capture or storage to cut peak draw. But where large data-centre clusters sit in water-stressed or infrastructure-constrained regions, where evaporative or hybrid cooling is used, where potable supplies are involved, and where drought and heatwave peaks coincide with farming and household demand, the **local cumulative risk is real**. Current UK, Irish and EU systems are improving on disclosure, but they are still poor at answering the public-interest question that matters most: **if the next hyperscale or AI campus is approved here, what happens to water availability, peak resilience, agriculture, public infrastructure and the wider local economy in this catchment, not just on this plot?** ⁵

Key findings

- The best quantified England-wide estimate located for this report is the Environment Agency's **underestimate of 1,125 ML/year** in 2024–25 for roughly 200 data centres with usable water information, with **937 ML/year** in Water Resources South East alone and clear summer peaking. ⁶
- Ireland's national utility has argued that data-centre water use is **less than 0.2% of total national demand**, but also acknowledges the need to manage **peak flows** on individual developments with storage and flow limits. That is not a contradiction; it is exactly the difference between national averages and local peak-system stress. ⁷
- The largest high-confidence European site figure found in public reporting for this study is **Meta Clonee**, with annual withdrawals ranging from **571 ML** in 2024 to **928 ML** in 2021. The lowest striking comparator is **Google Dublin** at about **2.7 ML** in 2023, reported after a shift to air cooling. ⁸
- Public official records also show substantial **permitted or requested** non-potable withdrawals even where actual use is not yet disclosed: **Equinix Slough** applied to abstract up to **1.728 ML/day** and **367.5 ML/year** from groundwater for evaporative cooling; **Amazon Data Services Spain** sought **204,697 m³/year** from the Canal Imperial de Aragón for a Zaragoza data centre, with a maximum instantaneous flow of **26.3 l/s**. ⁹
- The strongest regulatory improvement now under way is **public water reporting**. The weakest part remains **cumulative local assessment**, especially across clustered facilities and across drought and heatwave conditions. ¹⁰

Methodology and measurement issues

This paper prioritised primary and quasi-primary sources: government and regulator documents, water-company material, official planning and licence records, company sustainability disclosures, and peer-reviewed or specialist technical sources where they clarified definitions or hidden trade-offs. The strongest evidence in this report comes from the Environment Agency, Uisce Éireann, BOE water-rights notices, North Holland provincial strategy documents, Meta's environmental data index, and operator technical or sustainability pages from Google and Microsoft. Where a figure came through a secondary source, that is stated explicitly and the confidence has been lowered. ¹¹

Several measurement problems materially affect public understanding. First, **withdrawal** and **consumption** are not the same thing. Meta's methodology is a useful illustration: it defines water withdrawal as water taken from local utilities or aquifers and water consumption as water not returned, calculated either as withdrawal minus wastewater discharge or by cycles of concentration in cooling systems. Second, **potable** and **non-potable** sources are often mixed or not disclosed at site level. The EU's delegated reporting regulation now distinguishes **total water input** and **potable water input**, which should improve comparability, but only for the subset of facilities in scope and only where data quality is adequate. ¹²

Third, **annual averages hide peaks**. The Environment Agency found that England's data-centre water consumption peaks in summer and correlates with temperature, which is exactly when catchments, rivers, reservoirs, irrigation systems and water-treatment assets are often under the greatest pressure. Uisce Éireann's own explanation for Dublin likewise focuses on limiting peak flows and requiring private storage on large developments. That means annual numbers are necessary, but not sufficient, for real planning. ¹³

Fourth, **company data mixes measured values and estimates**. Meta states that where water activity data is incomplete or unavailable it may use existing activity data or publicly available intensity metrics as a proxy. That matters because even strong company reporters may still blend metered and estimated values. In practice, water figures should therefore be classified as one of the following: measured company-reported; utility-derived; planning-derived or licence-derived; regulator estimate; secondary estimate; or speculative. This report uses that distinction in the table below. ¹⁴

Finally, missing data is itself a finding. For some of the largest European hubs — especially Frankfurt, Paris and Milan — this research located abundant evidence on capacity growth, energy demand and policy debate, but very little reliable, site-level freshwater withdrawal or consumption data in the public domain. The absence is not neutral. It limits catchment planning, weakens comparison with other sectors, and advantages actors who can talk in global efficiency terms while keeping local trade-offs opaque. ¹⁵

What the known numbers show

The current evidence shows a sector with an extremely wide distribution of water use. Some facilities or campuses appear effectively low-water in direct operation; others are large enough to be material local users. The most important published contrast in Ireland is between Meta Clonee and Google Dublin. Meta's annual withdrawal at Clonee peaked at **928 ML** in 2021 and was still **571 ML** in 2024. By contrast, the An Fóram Uisce review of Google's reporting put Google Dublin at **2.7 million litres** in 2023 and explicitly linked that to an air-cooling configuration. This does **not** mean Google is globally low-water; it means that a specific site and a specific cooling design can be. ⁸

England's public evidence is better at regional scale than at facility scale. The Environment Agency's 2025 water-resources appendix says around **57%** of England's approximately **300** data centres are in the south-east, that **38 of the top 50 water-consuming data centres** are also there, and that observed consumption for about 200 centres reached **1,125 ML/year** in 2024–25, with **937 ML/year** in WRSE. The same appendix says those values are likely to be **underestimates** because data is missing for around 100 data centres. That combination — clustering, seasonal peaking and known undercounting — is precisely why cumulative local risk matters more than national averages. ¹⁶

In the Nordics, the public picture is different again. Meta's 2024 site disclosures show **29 ML** at Luleå and **292 ML** at Odense. Google's Hamina site uses seawater from the Gulf of Finland and links that to

off-site heat recovery. Microsoft’s Netherlands materials — useful as a near-Nordic comparator in cool-climate north-west Europe — say its newer Dutch facilities use outside air and zero water for cooling when temperatures are below **29.4°C**, reducing cooling-water use to less than **5% of the year**, while rainwater capture offsets humidification demand. These are not proof that cool-climate growth is impact-free, but they do show that cooler climates and access to alternative water sources can materially reduce **direct freshwater dependence**. ¹⁷

The table below consolidates the best public figures located for this report. It intentionally includes both **measured/disclosed values** and **official licensed or requested maxima**, because both matter. A licensed volume is not actual consumption, but it is the volume the system is being asked to accommodate. “Unknown” is used where no reliable value was identified. ¹⁸

Country	Region / Site	Operator	Facility type	Cooling type	Annual water use	Peak daily use	Potable water?
Ireland	Clonee, Co. Meath	Meta	Hyperscale campus	Unknown	571 ML withdrawal (2024)	unknown	unknown
Ireland	Dublin	Google	Hyperscale / cloud	Air cooling	2.7 ML withdrawal (2023)	unknown	unknown
Ireland	Greater Dublin Area	Mixed	Cluster	Mixed	unknown	unknown	unknown

Country	Region / Site	Operator	Facility type	Cooling type	Annual water use	Peak daily use	Potable water?
England	England total	Mixed	Mixed	Mixed	1,125 ML consumption (2024–25, estimated undercount)	unknown	unknown
England	WRSE region	Mixed	Mixed	Mixed	937 ML consumption (2024–25, estimated)	unknown	unknown
England	Slough, LD11	Equinix	Colocation / large campus site	Evaporative cooling	367.5 ML/year licensed abstraction limit	1.728 ML/day licensed max	Non-municipal ground source
Netherlands	Amsterdam / North Holland designated clusters	Mixed	Cluster	Mixed	unknown	unknown	unknown

Country	Region / Site	Operator	Facility type	Cooling type	Annual water use	Peak daily use	Potable water?
Netherlands	Middenmeer	Microsoft	Hyperscale / cloud	Direct evaporative + rainwater capture	unknown	unknown	Partly non-potable rainwater offset
Germany	Frankfurt / Rhine-Main	Mixed	Cluster	Mixed	unknown	unknown	unknown
Denmark	Odense	Meta	Hyperscale campus	Unknown	292 ML withdrawal (2024)	unknown	unknown
Sweden	Luleå	Meta	Hyperscale campus	Unknown	29 ML withdrawal (2024)	unknown	unknown
Finland	Hamina	Google	Hyperscale / cloud	Seawater cooling + off-site heat recovery	unknown	unknown	No freshwater figure disclosed; seawater source
Spain	Zaragoza / Canal Imperial de Aragón	Amazon Data Services Spain	Hyperscale / cloud	Unknown	204.697 ML/year requested water right	2.27 ML/day equivalent at 26.3 l/s max instantaneous flow	Non-potable canal water source

Country	Region / Site	Operator	Facility type	Cooling type	Annual water use	Peak daily use	Potable water?
Spain	Zaragoza / Aragón	Microsoft	AI-ready hyperscale region	Closed-loop, advanced air and liquid-to-chip cooling	unknown	unknown	Claimed water for cooling new site

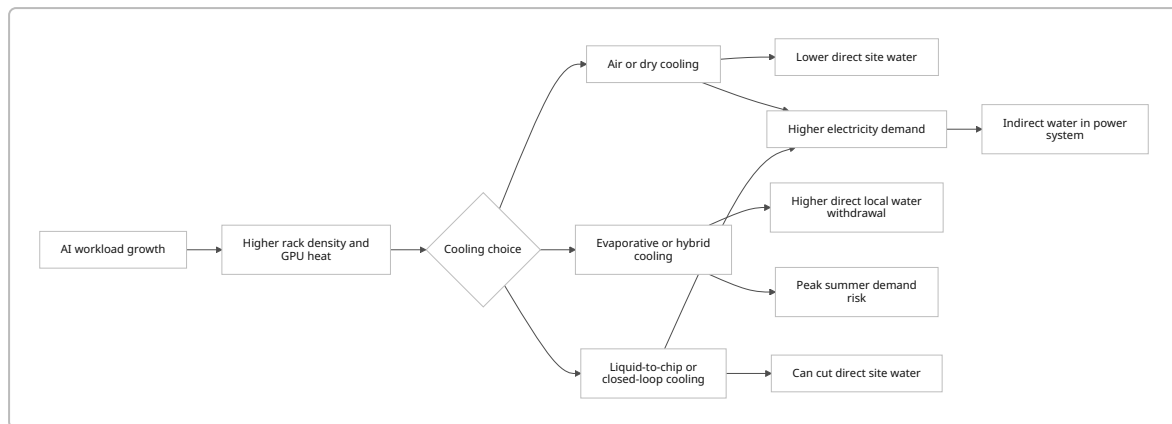
A simple way to read the European landscape is as a matrix of **hub density**, **local water stress**, and **disclosure quality** rather than as one continental average. Dublin and WRSE combine high concentration with already stressed or infrastructure-constrained systems and weak public site disclosure. North Holland combines high concentration with stronger spatial policy and establishment conditions. Frankfurt combines very high infrastructure density with better public debate on heat reuse than on water use. Aragón illustrates how southern European growth is colliding with explicit water-rights applications and drought sensitivity. Nordic examples show that cooler climates can lower direct freshwater demand, but do not remove the need for disclosure and local assessment. ²⁹

AI cooling and where the water goes

AI intensifies one part of the data-centre problem: **heat density**. More dense compute means more concentrated heat, faster thermal spikes, and greater pressure on cooling systems. But the direct water outcome depends on which cooling architecture is chosen. Google's climate-conscious cooling framework makes the trade-off explicit: in many locations, water cooling can be the most energy-efficient option, and Google says water-cooled data centres use about **10% less energy** and emit around **10% less carbon** than many air-cooled designs. That is environmentally relevant, but it also means that a planner or resident should not accept "low-carbon" language as a substitute for a water-accounting answer. ³⁰

Microsoft's current narrative shows the other side of the transition. It says new data centres designed for AI workloads can consume **zero water for cooling**, and that direct-to-chip cooling can save more than **125 million litres** per facility each year. In Spain, Microsoft says new Zaragoza facilities are being built with a **closed-loop** system using advanced air and liquid-to-chip cooling; in the Netherlands, it says outside air and direct evaporative systems reduce water use to less than **5% of the year** and that rainwater capture offsets part of demand; in Ireland it says rainwater harvesting is being used to offset water consumption at Irish data centres. These claims are credible as design strategies, but they remain mostly **operator claims** until matched with measured, site-level annual and peak-day reporting. ³¹

Google's own reporting, as summarised by An Fóram Uisce, also shows that AI can move the needle in the opposite direction. Google said the expansion of AI products and services increased data-centre workloads and the associated water footprint needed to cool them efficiently. That is consistent with a wider pattern: AI may reduce direct site water **in some new designs**, while increasing direct water **at other sites** and increasing **indirect** water in the electricity system when dry cooling or closed-loop liquid systems consume more power. Secondary reporting based on Lawrence Berkeley National Laboratory work suggests those indirect effects may be several times larger than direct cooling water, but hard European site-level evidence remains very limited. ³²



The practical answer to the report's AI question is therefore this: **AI can reduce, hide, relocate or intensify water use depending on the design and the metric used.** It reduces direct site water where closed-loop, air-cooled or liquid-to-chip systems work well. It hides water when companies publish fleet averages rather than site-level numbers. It relocates water when lower on-site use is bought at the cost of greater power-system demand. And it intensifies water risk when high-density campuses in warm periods rely on evaporative or hybrid systems drawing from already constrained local supplies. ³³

Regional case studies

Dublin and the Irish east

Dublin is the clearest example in these islands of why national averages can mislead. Uisce Éireann says data centres account for **less than 0.2%** of overall national demand, but the same utility also manages large users by limiting peak flows and requiring private storage. In the Greater Dublin Area, it reported domestic demand at **207 ML/day** and non-domestic demand at **139 ML/day** in 2019. In 2025 parliamentary debate, the Irish Government noted that the greater Dublin area remains highly dependent on the **River Liffey**, with **85%** of supply drawn from that source. A water user that is nationally small can still matter in a city-region dependent on a narrow source base and vulnerable peak conditions. ³⁴

Meta Clonee illustrates the scale that one campus can reach. Meta's own site disclosures show withdrawals of **615 ML** in 2020, **928 ML** in 2021, **839 ML** in 2022, **659 ML** in 2023 and **571 ML** in 2024. Spread evenly across a year, the 2021 figure equates to roughly **2.54 ML/day** and the 2024 figure to roughly **1.56 ML/day**. Google Dublin illustrates the opposite case: around **2.7 ML for the whole of 2023** after shifting to air cooling. The conservative lesson is that Dublin's public debate should move away from generic claims about "the sector" and towards **site-by-site disclosure of cooling mode, water source, and peak draw.** ⁸

Slough, London and the wider south-east

The south-east is where England's cumulative-risk problem is most visible in public data. The Environment Agency says around **57%** of England's data centres are in the south-east and that **38 of the top 50 water-consuming centres** are also there. For 2024–25 it estimated total England data-centre consumption at **1,125 ML/year**, of which **937 ML/year** sat in the already water-scarce WRSE region, and it found summer peaking in line with temperature. That is a direct warning that planning based only on annual averages will miss the stress point that matters most to utilities and catchments. ³⁵

Slough provides the clearest official site-specific water-rights signal located for this report. Equinix applied for a licence to abstract up to **72 m³/hour, 1,728 m³/day and 367,500 m³/year** from underground strata at London Data Centre 11 in Slough for **evaporative cooling**. That is not proof of actual use, but it is a formal statement of infrastructure demand. It sits in a region that the Environment Agency classifies as seriously water-stressed and in a wider Thames system that continues to manage drought and supply risk. The risk here is cumulative: multiple campus-scale facilities requesting water, electricity and land in one urban-industrial cluster, while water planning and planning permission often remain fragmented by site and by regulator. ³⁶

Amsterdam and North Holland

North Holland's response is notable because it explicitly acknowledges water as one of the sustainability limits that must shape where data centres can go. The province's data-centre strategy says the sector has a "huge impact" on landscape and uses "a lot of energy and water", limits new growth to designated business-park clusters in **Haarlemmermeer, Hollands Kroon and Amsterdam**, bans new data centres outside those areas, and makes establishment conditions depend on issues including **water use for cooling in relation to water availability, water safety and nuisance, and wastewater quality**. That is one of the clearest policy examples found in this research of water being embedded in siting strategy rather than treated as an afterthought. ²³

What remains weaker is public, site-level consumption disclosure. Microsoft's local materials show rainwater capture and very low annual cooling-water use in Dutch operations, which is important and positive. But North Holland's own strategy still relies on a mix of spatial controls, establishment conditions, permitting, monitoring and agreements with industry — suggesting that disclosure and enforcement are still evolving. Amsterdam's lesson for the UK and Ireland is not "ban them all"; it is **put water, energy, land and heat together in the siting conversation before clusters become locked in**. ³⁷

Frankfurt and Hesse

Hesse's official digital strategy openly identifies Frankfurt as Europe's data-centre capital and says the state hosts about **a quarter of Germany's total data-centre capacity**, while accounting for over **40%** of colocation and large-scale data-centre capacity. The strategy supports further growth, but it also says data centres account for a large share of total energy use and identifies more efficient cooling and waste-heat reuse as priority technologies. Frankfurt's public climate pages likewise name cooling systems for large data centres as sources of waste heat with urban potential. ²⁵

The problem is that public water data is much thinner than public heat-reuse rhetoric. This report did **not** locate robust, site-level Frankfurt water-withdrawal data comparable to Meta's Irish disclosures or the Slough and Zaragoza water-rights records. That itself is telling. Frankfurt's policy debate is more mature on **what to do with the heat** than on **what water the cooling systems require**, whether directly or indirectly. For planners elsewhere, Frankfurt's lesson is that heat-reuse ambition should not become a substitute for transparent water accounting. ³⁸

Aragón and the Spanish data-centre build-out

Southern Europe makes the climate question materially harder. Microsoft's Aragón material says Spain entered this phase after more than **three years of severe drought**, and presents its Zaragoza data-centre design as using a closed-loop system with advanced air and liquid-to-chip cooling to avoid water consumption for cooling at new sites. That is exactly the kind of design response a hotter climate demands. ³⁹

But the other side of the same regional story is the formal water-rights request. In 2025, the BOE published notice that **Amazon Data Services Spain** had applied to take water from the **Canal Imperial de Aragón** for a data centre in Zaragoza, with a **maximum annual volume of 204,697 m³** and a **maximum instantaneous flow of 26.3 l/s**. Those two facts together capture the real southern-Europe dilemma: climate and technology are pushing operators towards lower-water designs, while the scale of campus development still shows up in formal water claims on already sensitive systems. ²⁷

The Nordic comparison

The Nordic comparison is often used to imply that cooler climates solve the water problem. They clearly help, but they do not erase it. Meta's 2024 withdrawals of **29 ML** at Luleå and **292 ML** at Odense are far below high-water U.S. or Irish examples, yet they are not zero. Google's Hamina site is more interesting again because it uses seawater cooling and off-site heat recovery. In other words, the Nordic advantage is less "cold weather magically fixes it" than **a combination of cooler ambient conditions, alternative water sources, and stronger opportunities for heat recovery**. That still requires disclosure and local assessment, especially as AI raises rack densities even in cool climates. ⁴⁰

Local impacts on farming, food, utilities and economies

The most defensible conclusion here is that water conflict is usually **not** between data centres and agriculture in the abstract; it is between **all significant users at the wrong time in the wrong place**. Peak demand is the real fault line. The Environment Agency found England's data-centre water use peaks in warm months. Uisce Éireann's answers on Dublin focus repeatedly on peak-flow management and storage. North Holland's strategy explicitly links cooling-water use to water availability and wastewater quality. These are governance signals that the risk emerges under **heatwave and drought conditions**, not only in annual totals. ⁴¹

For scale, the 2021 **Meta Clonee** withdrawal of **928 ML/year** is equivalent to about **18,560 person-years** of domestic use using the UK benchmark of roughly **50,000 litres per person per year**. Its 2024 withdrawal of **571 ML/year** is about **11,420 person-years** on the same basis. The **Equinix Slough** licensed annual abstraction limit of **367.5 ML/year** equates to about **7,350 person-years**. The **Amazon Zaragoza** requested annual volume of **204.697 ML/year** equates to about **4,094 person-years**. These are not perfect comparisons — domestic demand differs by place, season and leakage, and industrial water is not interchangeable with household supply — but they give residents and councillors a more intuitive sense of magnitude. ⁴²

What about farming, food processing and ecosystems? The evidence is strongest at the level of plausible pressure pathways rather than quantified direct displacement. In Aragón, Microsoft is funding irrigation-efficiency projects in the Ebro basin designed to save **115,000 m³/year** by 2027 for farmers, which indicates the regional relevance of agricultural water efficiency in the same basin where new data-centre growth is occurring. In North Holland, the provincial framework directly links data-centre water use to water availability and wastewater quality. In Dublin, Uisce Éireann's own responses accept that large users must be managed around peak demand. The reasonable inference is that in stressed basins the real question is not whether one data centre "uses more than farming" in annual volume, but whether a cluster of facilities **hardens competition for resilience margins** that agriculture, food processing, river ecology and public supply all depend on in dry periods. ⁴³

The economic side is similarly asymmetric. Public authorities consistently recognise the economic value of digital infrastructure. Hesse wants to preserve Frankfurt's competitive lead. North Holland wants a strong basis for the digital economy while steering growth into designated areas. Microsoft argues that

its Spain cloud region will accelerate the AI economy and digitalisation. These benefits are real and should not be dismissed. But public reporting remains too weak to calculate robust **jobs per megalitre**, **tax per megalitre**, or **local value added per megalitre** across European sites. Planning documentation usually discloses either economic claims without water or water-system data without durable operational employment figures. That asymmetry benefits developers and larger operators because it makes economic upside legible while keeping water opportunity cost opaque. ⁴⁴

Regulatory gaps and cumulative impact

The EU framework is improving, but it is still not a catchment-allocation regime for AI infrastructure. The Water Framework Directive aims to halt deterioration and achieve good status for rivers, lakes, coastal waters and groundwater. The recast Energy Efficiency Directive, implemented in part through the 2024 delegated regulation, now requires owners and operators of in-scope data centres to make public sustainability information, including **total water input** and **potable water input**. That is an important step because it should create a comparable baseline that has not existed. ⁴⁵

Corporate reporting is also tightening. ESRS E3 under the European sustainability reporting standards covers surface water and groundwater and includes disclosure requirements about water consumption and related dependencies, impacts, risks and opportunities where material. That will improve board-level visibility and investor scrutiny. But these mechanisms are still mostly about **disclosure and transparency**, not direct permitting rules about who gets water first in a drought, how cumulative peak demand should be modelled at catchment scale, or when potable water use by data centres should be refused in favour of other users. The European Commission itself noted in 2026 that, beyond transparency and reputational considerations, the current framework does not set incentives for operators to improve sustainability performance nor measures to accelerate sustainable data centres and related investment. ⁴⁶

At national and regional level, the gap is even plainer. The Environment Agency says future non-household demand forecasts are unlikely to capture future data-centre needs because growth is so rapid and because too little is known about where centres will be built and what cooling methods they will use. Uisce Éireann works development by development and focuses on peak-flow limits and storage. North Holland's strategy is more advanced than most because it integrates energy, water, wastewater and spatial policy, but even there the emphasis is on conditions and cluster design rather than hard volumetric public-accountability rules for cumulative draw in one basin. My conclusion is therefore an evidence-based inference: **cumulative impact is the missing category** in much of present regulation. ⁴¹

Recommendations, final test and supporting material

Recommendations

The evidence collected here supports a proportionate reform agenda rather than a blanket ban.

Mandatory **site-level water disclosure** should apply to all large data centres and major expansions in the UK, Ireland and the EU, with separate reporting for annual **withdrawal**, annual **consumption**, annual **discharge**, **potable** input, **non-potable** input, and **peak-day** and **peak-hour** draw. The EU framework now has the bones of this through total and potable water input reporting; national planning systems should require the rest at development and operational stages. ⁴⁷

Planning and abstraction regimes should require **catchment-level cumulative assessment** for clustered development, not only site-by-site review. At minimum, this should test dry-year annual average demand, dry-year peak demand, heatwave scenarios, drought restrictions and interactions with public supply, irrigation, wastewater treatment and environmental flow obligations. The Environment Agency's own modelling appendix already demonstrates why this is needed. ⁴⁸

Where potable supply is proposed, decision-makers should require a formal alternatives test covering **reclaimed water, rainwater capture, storage, seawater where relevant, or lower-water cooling architectures**, and should require developers to explain why those options are not feasible. Google, Microsoft and AWS all now publicly advance variants of this practice; it should become a planning norm rather than a voluntary best-practice claim. ⁴⁹

Large schemes should disclose **public infrastructure costs** and funding responsibilities up front: network upgrades, water-treatment constraints, wastewater impacts, storage requirements, and any resilience works needed to protect households or ecosystems during drought. This is especially important where profits and high-value compute revenues are private, but local utility and environmental burdens are socialised. ⁵⁰

Operators should be required to publish **site-level cooling configuration** and any material changes to it, because cooling architecture is often the single biggest determinant of direct water demand. Without that, WUE numbers remain too abstract for public scrutiny. ⁵¹

Final test

Which claims are strongest?

The strongest claims are the ones grounded in official or directly disclosed numbers: Meta's site withdrawals; the Environment Agency's England estimate and seasonal peak pattern; Uisce Éireann's statements on national share and peak-flow management; Equinix's Slough abstraction application; and Amazon Data Services Spain's official water-rights request in Zaragoza. ⁵²

Which claims are weakest?

The weakest claims concern Europe-wide generalisations about what "AI data centres" use, claims that any one low-water design solves the wider problem, and estimates of indirect water use from electricity that are not yet routinely disclosed in Europe at site level. Those effects may be important, but the evidence base is still patchy and often secondary. ⁵³

Where is the data missing?

It is most obviously missing in site-level figures for major European hubs such as Frankfurt, Paris and Milan; in **peak-day** and **peak-hour** demand; in potable versus non-potable splits; in wastewater discharge volumes and temperatures; in indirect electricity-related water use; and in public, comparable links between water use and long-run local employment or tax contribution. ¹⁵

Who benefits from opacity?

Operators and, sometimes, permissive planning systems benefit most. Operators can emphasise fleet-wide efficiency gains or water-positive goals while limiting scrutiny of site-specific burdens. Regulators and planning authorities also benefit from opacity because fragmented disclosure avoids politically difficult allocation decisions between homes, industry, farms and ecosystems in dry periods. ⁵⁴

Who carries the local risk?

Households dependent on already constrained systems, farmers and food processors reliant on

drought resilience, utilities that must manage peak supply, wastewater networks handling unusual return flows, and ecosystems dependent on river flow and groundwater status carry the risk. In practice, local government often carries the political risk, while catchments carry the physical one. 55

What should a planner, farmer, journalist or resident know before approving another major data centre?

They should know the site’s **annual withdrawal, annual consumption, potable share, non-potable share, peak-day demand, cooling system, backup source, wastewater discharge profile**, and **catchment-level cumulative demand including other planned sites**. They should also know what happens under a drought order or heatwave: who is curtailed first, who pays for upgrades, and whether the business case still stands if non-potable water is required. 56

What single disclosure would most improve public understanding?

A public, site-level disclosure of **annual and peak-day water withdrawal and consumption, split by potable and non-potable source, with cooling mode stated**. Nothing else would improve comparison so quickly. 57

What regulatory change would most improve accountability?

A requirement that planning and water regulators conduct a **joint cumulative catchment assessment** for any large data-centre scheme or cluster, covering drought and heatwave scenarios and publishing the result before consent. That would align the project boundary with the real boundary of water risk. This is an inference from the evidence, but it is strongly supported by the Environment Agency’s admission that current forecasts cannot accurately predict future sector demand without better data. 58

Open questions and limitations

This report did not locate a comprehensive public register of site-level data-centre water demand across the UK, Ireland and the EU. It also did not locate robust, comparable public datasets linking water use to permanent operational employment, local tax yield or public infrastructure cost. For some hubs, especially Frankfurt, Paris and Milan, the public debate is much richer on energy and land than on water. Those gaps are substantive findings, not minor research inconveniences.

Full source table

Title	Author / organisation	Date	URL	Accessed
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Appendix of calculations

Domestic-use benchmark used for transparent equivalence calculations

This report uses the figure quoted in the UK parliamentary briefing that average use is roughly **137 litres per person per day**, about **50,000 litres per person per year**. Because the source wording is person-based rather than household-based, all equivalence calculations below are expressed as **person-years of domestic use**, not households. ⁵⁹

Meta Clonee, 2021 daily average

928 ML/year = 928,000,000 litres/year

$928,000,000 \div 365 = \mathbf{2,542,466 \text{ litres/day}} \approx \mathbf{2.54 \text{ ML/day}}$. ¹⁹

Meta Clonee, 2024 daily average

571 ML/year = 571,000,000 litres/year

$571,000,000 \div 365 = \mathbf{1,564,384 \text{ litres/day}} \approx \mathbf{1.56 \text{ ML/day}}$. ¹⁹

Meta Clonee, 2021 domestic-use equivalent

$928,000,000 \div 50,000 = \mathbf{18,560 \text{ person-years}}$. ⁶⁰

Meta Clonee, 2024 domestic-use equivalent

$571,000,000 \div 50,000 = \mathbf{11,420 \text{ person-years}}$. ⁶⁰

Equinix Slough licensed annual limit domestic-use equivalent

367.5 ML/year = 367,500,000 litres/year

$367,500,000 \div 50,000 = \mathbf{7,350 \text{ person-years}}$. ⁶¹

Amazon Zaragoza requested annual volume domestic-use equivalent

204.697 ML/year = 204,697,000 litres/year

$204,697,000 \div 50,000 = \mathbf{4,093.94 \text{ person-years}} \approx \mathbf{4,094 \text{ person-years}}$. ⁶²

Amazon Zaragoza max-flow daily equivalent

$26.3 \text{ litres/second} \times 86,400 \text{ seconds/day} = \mathbf{2,272,320 \text{ litres/day}}$

$= \mathbf{2,272.32 \text{ m}^3/\text{day}}$

$= \mathbf{2.27 \text{ ML/day}}$ if sustained continuously. This is a hydraulic maximum-equivalent, not a reported actual operating day. ²⁷

Bottom line

The public will understand this issue far better when every major data centre publishes a simple, comparable operational water sheet: annual withdrawal, annual consumption, potable share, non-potable share, discharge, peak day, cooling mode, and catchment. Until that exists, arguments about whether AI infrastructure is “water-light” or “water-heavy” will remain too easy to weaponise, and too hard to verify. ⁶³

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